Daniel Kim

608-422-2004 | linkedin.com/in/daniel-kim/ | ddkim22@wisc.edu

Education

University of Wisconsin - Madison **Sep 2023 – Est. May 2027**

- Electrical Engineering: Semiconductor Engineering, Bachelor of Science *Madison, WI*
- GPA: 3.62/4.0 | 4x College of Engineering Dean's List
- Undergraduate Teaching Assistant (ECE 203: Signals, Information, and Computation, Fall 2026)

Research Experience

Wisconsin Electric Machines and Power Electronics Consortium (WEMPEC) **May 2025 - Present**

Research Intern / Advisor: Prof. Chirag Gupta *Madison, WI*

- Operated **wedge bonder** equipment for device packaging; successfully completed **50+ wire bonds per week** with **>95% yield**; trained 3 graduate students, reducing operating time by **40%.**
- Performed **Transfer Length Method (TLM)** based electrical measurements to characterize **MOSFET** devices, achieving **4.5 Ω·mm contact resistance** and **2.8 kΩ/sq sheet resistance**.
- Supported **cleanroom operations**, including wafer cleaving, photolithography preparation, resist coating, chemical handling, and vapor cutting for semiconductor device fabrication.

Leadership Experience

Korean-American Scientists and Engineers Association (KSEA)

National Board - *Director* **May 2025 – Present**

- Advised **37 KSEA Young Generation chapters** nationwide on member recruitment, funding strategy, and cross-chapter communication, strengthening operational stability and leadership effectiveness.
- Organized **national-scale leadership conferences** (Katalyst, IMPACTs, Young Generations Forum), providing platforms for mentorship, professional development, and networking for **300+ Korean-American** students and young professionals in STEM.

Wisconsin Young Generation Group - *President / Academic Advisor* **Jan 2024 – Present**

- **Doubled** the number of organization members from approximately **30 to 60+** by implementing academic programs including career-related guest lectures, research paper analysis, debates, mentoring programs with graduate students and faculty members.
- Provided project oversight, offering guidance and support to ensure successful completion of multiple initiatives.

Projects

AI-Agent Productivity Platform **Jan 2026 – Present**

- Developed a **full-stack system** enabling **multi-model agent collaboration**, content verification, and **style-aware writing generation**, with a **Next.js** frontend and **Spring Boot/FastAPI** backend services.
- Implemented backend AI workflows and system logic by separating product services and AI execution, supporting agent debate, search-and-verify pipelines, and structured run tracking using **PostgreSQL (JSONB)**.

Segway Hoverboard Digital Control System | *System Verilog, Synopsys Design Compiler* **Aug 2025 – Dec 2025**

- Built an FPGA-based **real-time control SoC** in **System Verilog**, integrating multi-sensor **gyro/accelerometer fusion**, **SPI/UART/A2D** interfaces, and **PWM** motor outputs with fixed-point **PID** control.
- Achieved **timing closure at 333MHz** by pipelining critical data paths and applying synthesis constraints in **Synopsys Design Compiler (SAED 32nm),** validated through module/full-chip testbenches and post-synthesis timing analysis.

Digital Assisted Smart Intervention App **Mar 2025 – Nov 2025**

- Developed a rehabilitation prototype using **real-time camera input** with an **angle-detection algorithm** and human-shaped motion overlay, providing **color-coded feedback** to guide correct posture and exercise execution.
- Designed the user interface in **Figma** and integrated **motion detection**, demonstrating the feasibility of **remote, posture-guided rehabilitation** interventions with ongoing validation against stored exercise gestures.

Smart Night Lamp **Jul 2024 – Dec 2024**

- Built and deployed a motion-activated smart lamp using **Arduino Nano**, PIR/LDR sensing, PWM dimming, and a custom **EasyEDA** PCB, implementing embedded control logic that cut standby power to **<0.1 W** and delivered **~40% energy savings** while improving campus safety.

Research Publications

Observation of Impurity Band Transport in Al-rich and n-AlGaN films using Low-temperature MOCVD

Mukhopadhyay S. *et al.*, **Kim, D. ,**University of Wisconsin Madison, Department of Electrical and Computer Engineering, 2026

Skills

Software: Java, MATLAB, Assembly, System Verilog, LTspice, Arduino, EasyEDA

Technical Skills: Analog & Digital Circuit Design, Signal Processing, Soldering Techniques, TLM Measurements, Wedge Bonding